ROSHAN ROY VARGHESE

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

MSc Artificial Intelligence student (Heriot-Watt University, distinction-level dissertation) with a full-stack development background. Built a production-grade agentic RAG system achieving 100% source accuracy over 753 pages of regulation, and an NLP evaluation pipeline processing 600 LLM outputs with statistically validated results. Proficient in Python, React, FastAPI, and modern ML tooling. Proven ability to ship client-facing software and deliver under pressure.

EDUCATION

Heriot-Watt University — MSc in Artificial Intelligence Sept 2025 – Sept 2026 (expected)
Edinburgh, Scotland

- Focus: Natural Language Processing (NLP), Large Language Models (LLMs), Explainable AI (XAI). Dissertation scored 84/100 (distinction).

Amal Jyothi College of Engineering — BTech in Computer Science June 2020 – May 2024
Kerala, India

- GPA: 6.83. Selected for Smart India Hackathon 2023 (national-level, campus representative).

EXPERIENCE

Freelance Web Developer 2023 – Present

- Designed and delivered client websites including a bilingual English/Arabic e-commerce catalogue for a premium Omani brand (Suhool Shinas INTL), with RTL layout switching, WhatsApp ordering integration, and glassmorphism UI.
- Built responsive multi-page sites using HTML, CSS, JavaScript, React, and Tailwind CSS, deployed via Netlify.

McDonald's — Customer Care Assistant Oct 2025 – Present
Edinburgh, Scotland

- Awarded Employee of the Month (February 2026) for exceptional teamwork, communication, and customer service.
- Collaborated with shift management in a high-volume environment, developing strong interpersonal skills and ability to perform under pressure.

Acmegrade — Web Development Intern Aug 2023 – Sept 2023

- Developed responsive web applications using HTML, CSS, and JavaScript. Implemented mobile-first design principles across multiple projects.

PROJECTS

Agent Runtime Platform
LangGraph, Docker SDK, Pydantic, FastAPI, React · In Progress

- Building a self-correcting data-analysis agent: given a CSV and a question, it plans the analysis, generates Python, executes it in a sandboxed Docker container, and retries on error to return a cited report.
- Architected around a LangGraph plan-execute-evaluate-retry state machine with Pydantic-typed state, structured so compliance, repository, and research workflows can be added without changing the runtime core.

Compliance RAG Assistant
Python, FastAPI, ChromaDB, React, LLMs | Live Demo

- Built a production-grade agentic RAG system answering questions over GDPR, EU AI Act, and UK Data Protection Act (753 pages, 5,636 indexed chunks), grounding every answer in cited source text to address hallucination and traceability gaps in LLM adoption for regulated industries.
- Implemented hybrid retrieval combining dense vector search and BM25 keyword search with Reciprocal Rank Fusion, followed by cross-encoder reranking (ms-marco-MiniLM), achieving ***100% source retrieval accuracy and 98.6% faithfulness*** on a 10-question evaluation set (LLM-as-judge methodology).
- Designed an agentic routing layer where the LLM classifies queries and dispatches to document-specific search, full-corpus search, or cross-regulation comparison, using metadata filtering to eliminate retrieval noise.
- Developed a full-stack application (FastAPI backend, React frontend) with source citation display, PDF viewer, and Docker containerisation.

Investigating the Faithfulness–Simplicity Trade-off in Generative Adaptive XAI
MSc Dissertation, Python, SHAP, LIME, XGBoost, Hugging Face, Groq API

- Built an end-to-end NLP evaluation pipeline: trained XGBoost classifiers (ROC-AUC 0.799 / 0.927), generated 600 adaptive explanations via Llama-3.3-70B (Groq API), and created a novel "Textual ShapGAP" metric to quantify faithfulness in free-text explanations.

- Demonstrated a statistically significant faithfulness-simplicity trade-off (paired t-test: $t(149) = 4.88$, $p < 0.001$) while disproving length as a confound (Spearman $\rho = -0.02$). Conducted an ethics-approved human evaluation study ($n = 10$). Scored 84/100.

Suhool Shinas INTL — E-commerce Catalogue
HTML, CSS, JavaScript, Netlify | *Live Site*

- Fully responsive bilingual (English/Arabic) multi-page e-commerce catalogue for a premium Omani heritage clothing brand with worldwide delivery.
- Implemented RTL layout switching, dynamic product catalogue from JSON, WhatsApp-integrated ordering, image lightboxes, PDF catalogue viewer, and cart sidebar with badge counter.

UrbEstate | Kobodrop Clone | FIT Fitness Website
React, Tailwind CSS, Vite, HTML, CSS, JavaScript

- Portfolio of responsive web applications: real estate listings platform with dynamic filtering and animations, file-sharing clone with drag-and-drop uploads, and a mobile-first fitness website.

TECHNOLOGIES

- **AI / ML:** NLP, LLMs, RAG, Explainable AI (XAI), SHAP, LIME, XGBoost, Word2Vec, Model Fine-Tuning, Hugging Face, sentence-transformers
- **Frontend:** HTML, CSS, JavaScript, React, Tailwind CSS, Vite, TypeScript
- **Backend:** Python, FastAPI, Node.js, SQL, ChromaDB, Langfuse
- **Cloud / DevOps:** Docker, Git, GitHub, GCP (certified), Oracle Cloud (certified), Azure